CHAPTER XXXVII

ON THE NOTION OF 'SIMULTANEITY'

> So we see that we cannot attach any *absolute* signification to the concept of simultaneity, but that two events which, viewed from a system of co-ordinates, are simultaneous, can no longer be looked upon as simultaneous events when envisaged from a system which is in motion relatively to that system. (155)
>
> A. EINSTEIN

In the older days we accepted as self-evident the structural assumption that there is sense in such a statement as that an event A on the sun was 'simultaneous' with an event B on the earth. We assumed also that the 'moments of our consciousness' had a universal 'meaning'. We tacitly assumed, for instance, that when we saw or photographed an event on the sun, that it happened just the moment we saw it. Such structural assumptions were rudely disturbed by the discovery of the *finite* velocity of light. Today we know that when we see or photograph an event on the sun, that event happened approximately eight minutes earlier, as it takes about eight minutes for the light from the sun to reach our earth. We begin to realize that the moments of our perceptions have no universal significance.

We inquire first what we mean structurally by simultaneity. We do not need to go into details. The application of functional and contact methods, even in the rough, will assist us. We can speak in terms of instruments. For instance, we can build a special, very fast moving picture camera, C, with two lenses D and E, at two opposite sides, and a calibrated film, F, running rapidly through the middle of the camera as shown in Fig. 1. If we focus our double camera on two flashes, A and B, occurring at 'equal distances', *L*, from the film, we say that the flashes occur simultaneously by definition if the pictures, a and b of the flashes A and B, appear exactly opposite each other on the film, or if we have *one* picture. If, under the conditions of the experiment, where the distances between the origins of the flashes and the film are equal, and our film is moving very rapidly, the pictures of the flashes do not occur exactly opposite each other, but one picture is separated from the other, then we have two pictures, and conclude, by *definition*, that the flashes are *not* simultaneous.

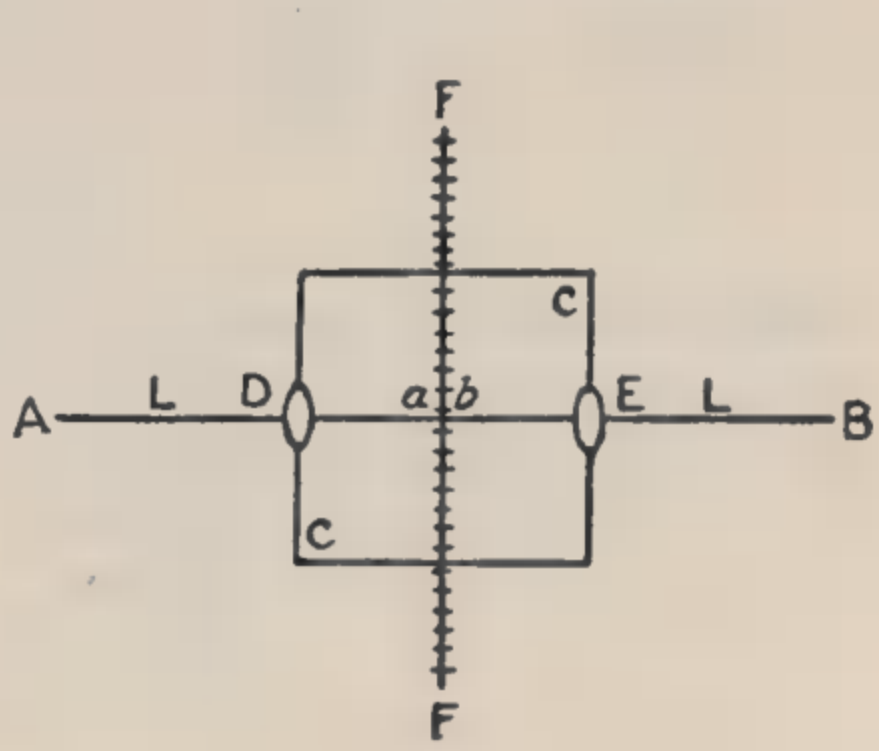

FIG. 1

We introduce this hypothetical instrument to show that, in discussing physics, and the theory of Einstein in physics, we do not speak of 'psychology' or personal 'subjectivity', but that we do deal with the inherent physical sub-

jectivity of the instruments and the finite velocity of propagation. When we discuss the psycho-logical or methodological or semantic significance of science and scientific method, we deal with different subjects.

When we use the term 'observer' we mean an observer so equipped that he can do whatever is demanded of him.

What was said about the definition of 'simultaneity' by the aid of the camera, applies also to ourselves.

The problem of prime importance before us is to find out if 'simultaneity', as defined, has an 'absolute' and universal significance, or if it is perhaps a private and relative notion.

We will carry out the analysis in two ways, the first by example, which will be instructive, though perhaps not completely conclusive; the other, by the use of the Lorentz-Einstein transformation.

Let us perform our last experiment, which, with modern methods seems to be feasible, in a slightly more complicated form.

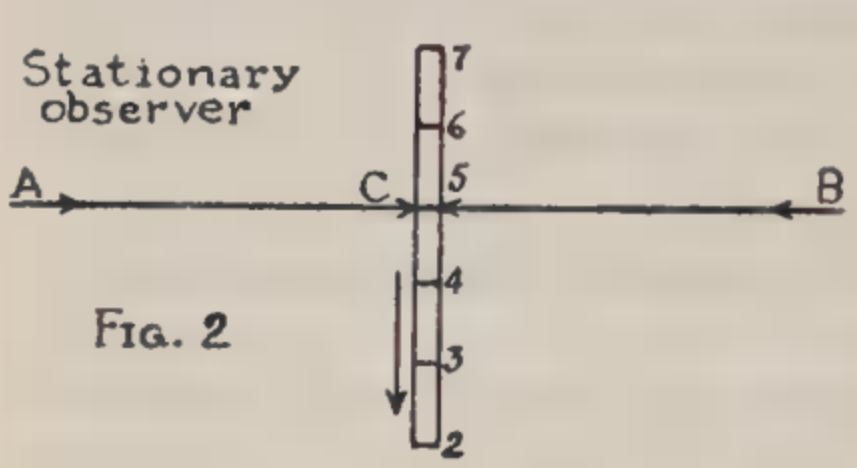

FIG. 2

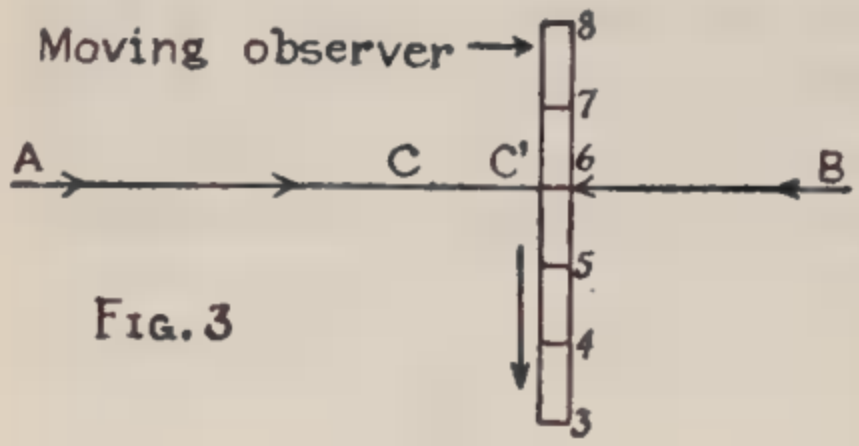

FIG. 3

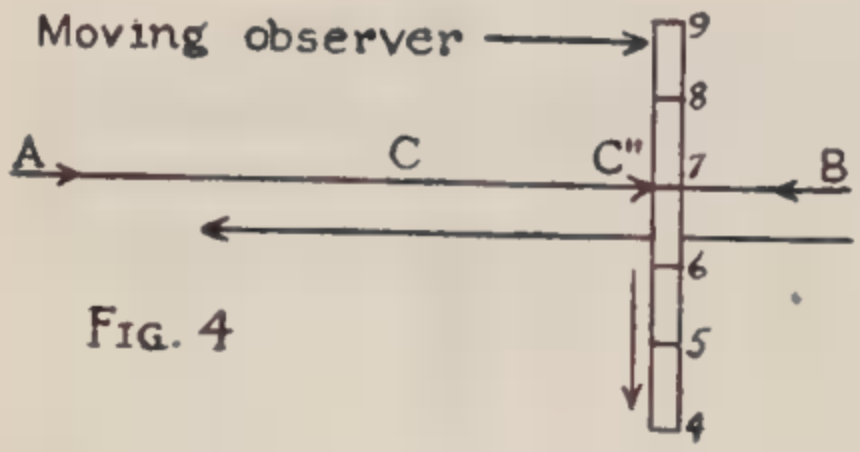

FIG. 4

We can select a dark night in which flashes will photograph well even at considerable distances. We can place powerful projectors at A and B, and we place our camera so that the film will come exactly at C, midway between A and B. We can start the mechanism of the rapidly moving film and, by an electrical contact made at C, we can produce a short flash from each of the two projectors. Because of the assumptions, AC = CB, and equal velocity of the propagation of electrical currents and light-waves in all directions we shall have by the structural definitions which condition the experiment, *one* picture in Fig. 2, say at the spot of our moving film marked by 5. The rays of light from A and B would arrive 'simultaneously'—that is, 'at the same time'—and would affect our moving film in *one spot*. Our definition was for a stationary observer, and under the conditions, the experiment was fairly definite—all the underlying structural assumptions, of course, being taken for granted.

Now consider an observer, as shown in Fig. 3, moving uniformly in the direction from A to B.

Let us assume that he is also equipped with a similar sort of moving picture camera as the stationary observer, and that just before he passes the point C

the electrical impulse to the projectors is sent. Let us assume further that the mark 5 on his moving film is exactly at the focal point of the camera as C is passed. The electrical impulses travelling from C to A and B would travel the distance AC = BC, produce the flashes A and B which again would travel with finite velocity in all directions. During the interval these impulses and light-waves are travelling, our observer is moving from A toward B, and spot 5 on his moving film is no more at the focus of the camera. Obviously he will meet the light-wave from B first, at C′, let us say, when mark 6 on his film is at the focus (Fig. 3). After another short interval when he reaches C″ and mark 7 on his film is at the focus, the light-wave from A overtakes him (Fig. 4).

So we see that what was 'simultaneous' (by definition) and produced *one* impression on the moving film of the stationary observer, was not 'simultaneous', (again by definition), for the moving observer, as *his* film registers *two* pictures.

As both observers use similar instruments and one set of definitions, obviously both are entitled to claim that their records on the film are conclusive. So the first can claim that the flashes were 'simultaneous', the second can claim that they were not 'simultaneous'. The reverse is equally true. If the moving observer had *one* picture, and claimed 'simultaneity', the stationary observer would have *two* pictures, and deny 'simultaneity'.

But when two observers are *equally justified* in making *two* opposing claims where, by their very meanings, there is only one possible, we must conclude that the claim itself is meaningless. We see that 'absolute simultaneity' is a fiction and impossible to ascertain, as it would depend on some impossible 'absolute motion', or 'infinite velocity' of propagation of signals.

The analytical form of showing the impossibility of 'absolute simultaneity' is very simple, and follows directly from the Lorentz-Einstein transformation.

Let us imagine two observers, one in an S system of co-ordinates (x, y, z, t) and another in an S' system of co-ordinates (x', y', z', t') moving relatively with the velocity v.

Let us assume two events happening in the unprimed system at the point (x_1, y_1, z_1) at the 'time' t_1, and the other at the point (x_2, y_2, z_2) at the 'time' t_2. According to the Lorentz-Einstein transformation the 'times' at which the two events occur relatively to the primed system are given by the formulae:

$$t_1' = \beta\,(t_1 - x_1 v/c^2),\quad t_2' = \beta\,(t_2 - x_2 v/c^2),\quad \text{where as usual } \beta = 1/\sqrt{1 - v^2/c^2}.$$

If we assume that in our unprimed system S the two events were 'simultaneous', which means that they 'occurred at the same time', t_1 would be equal to t_2, that is, $t_1 = t_2$, or $t_1 - t_2 = 0$. Let us find the difference between the two primed 'times' in the moving system S', and see if this difference is zero, which would mean that the primed 'times' are equal.

Returning to our formulae which give us the values for the primed system 'times', we express their difference as

$$t_1' - t_2' = \beta\,(t_1 - x_1 v/c^2) - \beta\,(t_2 - x_2 v/c^2) = \beta\,(t_1 - t_2 + x_2 v/c^2 - x_1 v/c^2). \qquad (1)$$

But we assumed $t_1 - t_2 = 0$; therefore $t_1' - t_2' = \beta\,(x_2 v/c^2 - x_1 v/c^2)$.

This last formula shows clearly that $t_1' - t_2'$ cannot be zero; or in other words, t_1' cannot be equal to t_2' unless $x_1 = x_2$.

The two events which, for an observer in the unprimed system, happen 'simultaneously', ($t_1 = t_2$, or $t_1 - t_2 = 0$) at different places, (or x_1 not equal to x_2, $x_1 \neq x_2$), cannot be 'simultaneous' for the moving observer in the primed system S', but will happen at different 'times' (t_1' is not equal to t_2', or $t_1' - t_2' \neq 0$).

It is extremely instructive to consider further what happens in measuring 'times' and 'lengths' in systems which are moving relatively to each other.

If, in the equation (1) above, we assume $x_1 = x_2$, this means that both events occur at one place in the stationary unprimed system S.

By changing the signs and cancelling the terms with x_1 and x_2, which are equal and of opposite signs, we have $t_2' - t_1' = \beta\ (t_2 - t_1)$ whence, substituting for β its value $1/\sqrt{1 - v^2/c^2}$, we obtain

$$t_2' - t_1' = \frac{t_2 - t_1}{\sqrt{1 - v^2/c^2}}$$

This last formula brings out a few remarkable issues. In terrestrial velocities the square of the velocity of the motion v^2 of the observer in the primed system S' is very small as compared with the square of the velocity of light c^2, so the fraction v^2/c^2 is small, $\sqrt{1 - v^2/c^2}$ differs very little from unity but the whole denominator is *less* than unity, and so $t_2' - t_1'$ is *not* equal to $t_2 - t_1$, but greater.

In other words, the interval of 'time' between the two events appears *larger* to the observer in the primed moving system than to the observer in the stationary unprimed system. In general, among all systems in a state of uniform relative motion, that one in which two events occur at *one place*, is characterized by the fact that the 'time' interval between the two events appears *shortest* to an observer in this system. The shortest interval means that to an observer in the system, the events run their course most rapidly. A process which, with reference to a given system, occurs in *one* place, appears to run its course most rapidly to an observer in that system, but more slowly to a moving observer in any other system.

The more rapid the relative motion, the slower the process will appear, and, in the limit, if an observer could move with the velocity of light, $v^2 = c^2$, the denominator of our equation would become $1 — 1 = 0$ and $t_2' - t_1'$ would become 'infinite' and all events would be at a standstill.

As the formulae for length, x and x', involve the 'times' and, as we see, the intervals of 'time' are dependent on the relative velocities, by a similar process of reasoning we find that the standards of length are also relative, and that the length L' in the primed system is represented by $L' = L\sqrt{1 - v^2/c^2}$. In other words, to an observer who sees the rod in motion, it will appear 'shortened', and among all systems in a state of uniform relative motion, the one in which the rod is at rest is distinguished from all others by the fact that in it the rod appears longer than in any other system. For instance, a metre rod lying on the earth in the direction of its motion would appear to an observer on the sun to be shortened by 5×10^{-7} cm. In the limit, when $v = c$, the fraction $v^2/c^2 = 1$, $1 - 1 = 0$ and $L' = 0$, which means that to an observer moving with the velocity of light, a three-dimensional body would appear as two-dimensional,

or a two-dimensional figure as one-dimensional. The co-ordinates y and z, as we have seen, do not enter into consideration as they are equal in both systems moving relatively in the X direction, and the 'time' co-ordinates are independent of them.

If a body at rest appears to the observer in the unprimed system as a sphere, it will appear as an *oblate spheroid* to an observer in the primed system.

We see that structurally not only 'simultaneity' and 'time' are not absolute but also that length, and therefore *shape*, is relative.

We have seen that the 'shortest' and 'longest' values are important characteristics of the motion. This suggests why in the general theory of Einstein we are interested in, and introduce, geodetics.

It should be mentioned here that the Lorentz transformation has been reached by difficult considerations involving Maxwell's electromagnetic field equations, unrelated to the Einstein theory. Einstein found the Lorentz transformation by the *simplest consideration* connected with his theory. The finding of such important equations by two methods, entirely different structurally, must be considered as a convincing proof of the fundamental importance of such formulae, the more so since they follow from very simple and fundamental structural principles which in themselves cannot be denied because they are negative in character. Negative statements are on a different footing in the new systems; they follow structurally from a $\bar{A}$ orientation, just as the older positive dogmas were the structural results of aristotelianism and the delusional results of identification.

The facts mentioned concerning the measures of length and the behaviour of clocks do not present any paradoxes. They simply say that these discrepancies are mutual and inevitable, as any measurement is only a measurement when it can be registered by an instrument, or seen, or recorded in some way. If the measuring rods and clocks are moving relatively to us, what we see or what our instruments record is *not* what is happening on the moving system, which no one can see or record from outside the system. What reaches us is simply what the light-waves or other signals moving with finite velocity (and therefore retarded by a motion away from us) bring to us. As all existing methods of communication and all known signals have *finite velocities*, these structural differences which are conditioned by the inherent characteristics of the world should be taken into consideration in modern science.

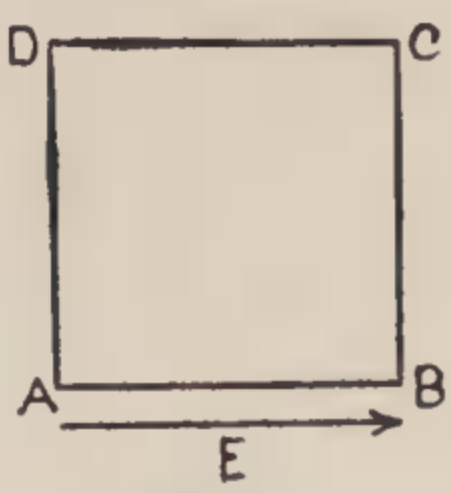

FIG. 5

If we draw a square $ABCD$ (Fig. 5) and an aviator E were to pass this square sign with a velocity of 161,000 miles a second* in the direction AB,

*I deliberately select such a velocity so as to make the contraction given by the formula $L' = L\sqrt{1 - v^2/c^2} = 1/2$. With this aim we must make the fraction represented by $v^2/c^2 = 3/4$, then $1 - 3/4 = 1/4$ and $\sqrt{1/4} = 1/2$. We find the square of our velocity v by taking 3/4 of the square of the velocity of light $v^2 = \frac{3}{4}c^2$ and find $v = \frac{c}{2}\sqrt{3} = 161{,}000$ miles a second.

he would see—and any instrument carried by him would register it—the sides of *our square* ($AB=BC=CD=DA$) in the direction of his flight, namely AB and CD, as 'contracted' to half their length. If he turned at right angles, the sides AB and CD would 'expand' and the other sides, which are at right angles, BC and DA, would 'contract'. For us the sides AB and BC are *equal*, for him one appears twice the other. To him our square appears oblong.

Under such *natural structural* conditions it is a fundamental fallacy to ascribe to 'lengths' or 'shapes' or 'times' any 'absolute' significance. If we grasp the structural fact that 'length' and 'duration' are not *things* inherent in the external world, nor are 'matter', 'space', and 'time', but that they appear as relations between events and some specified observer, and forms of representations, then all paradoxes would disappear.

A suggestion which concerns visualization may be helpful. If we realize the structural fact that words *are not* the objects they represent, we shall always discriminate automatically between what we *see*, *feel* . , on the level of lower order abstractions, and what we *say* on the level of higher order abstractions. When we have conquered that single difficulty we could never then identify the two different orders of abstractions. We would evaluate the *terms* 'matter', 'space', and 'time' as forms of representation, and non-objects, and we would describe events in a functional, operational, behaviouristic language of order. If we realize and feel the *finite* velocity of propagation of all processes, we may visualize all that has been explained here. Diagrammatizing and even following with one's hand, the *visualized order* of occurrences, helps enormously. Try to visualize how the aviator in the last example is flying away and how much more slowly the light impressions from the earth are reaching him or his instruments, and the difficulties will soon vanish.

We shall also be greatly helped in our power of visualization when we become acquainted with the structure of the Minkowski four-dimensional world. An explanation of this appears in the next chapter.

CHAPTER XXXVIII

ON THE 'WORLD' OF MINKOWSKI

> Moreover, the really fundamental things have a way of appearing to be simple once they have been stated by a genius, who was in this case Minkowski. (431)
>
> G. Y. RAINICH

We have already freely used the structural term 'dimension' and only hinted at its meanings. Before we approach the Minkowski world we must summarize roughly what for our purpose we should know about dimensions.

There is nothing mysterious about the term 'dimension'. First of all, the dimensionality of a manifold is not inherent in the manifold as such. It is a characteristic of *order* and so of structure. A manifold can be ordered in different ways, so that it follows that one manifold may have different dimensionality, depending on how we *order* it.

A manifold which has linear order and structure is called one-dimensional. A two-dimensional manifold is then a linearly ordered manifold of linearly ordered manifolds , .

Usually we speak about our 'space' of daily experiences as a three-dimensional manifold, but this is true only with reference to points, and not true with reference to lines or spheres. The manifold of all spheres in 'space' is, for instance, a four-dimensional manifold; so also is a manifold of lines.

Let us explain the line-dimensionality of our 'space' in terms of lines. A line can be given by two points—one, let us say, in the floor of our room, the other in the ceiling. Each of these points is given by two co-ordinates; it has two degrees of freedom; and so our 'space' is a four-dimensional (2×2) manifold *in lines*. This means that to distinguish any line in our 'space' from any other line we would have to have *four data*. Similarly, if we deal with spheres, a manifold made up of spheres requires four data, three for locating the centre and one giving the radius of the sphere. The above examples, of course, do not exhaust the structural possibilities.[1]

The term 'dimension' does not apply solely to what we call 'space'. The term applies to any manifold which we can order in some particular way. Manifolds or aggregates abound everywhere in our lives. The domain of colours, for instance, is a manifold; and so is the domain of tone, or of remembrances , . No manifold in itself has any dimensionality. To ascribe dimensionality to the manifold we must first order it and the number of its dimensionality, or its ascribed or discovered structure, may differ according to the principle of ordering used.

In discussing dimensionality we have two purposes. First, to dispel the semantic fright about this simple term; and, second, to suggest *means for visualization*, which for our purpose are of great neurological importance.

When we say that the world is structurally a four-dimensional manifold, we mean only that according to our experience and the structure of our nervous

system, the world of our experience is represented by a fourfold order. We can order the events as to the right and to the left, forward and backward, up and down, and sooner and later. In our experience this fourfold order is completely united, and cannot be separated unless we deliberately choose to *neglect* some of these orderings.

Nor does it mean that all these dimensions are 'identical'. We are accustomed, for instance, to consider the three dimensions of 'space' as 'identical', or at least equivalent. Is this true in life? Can we disregard, for instance, the structural difference between vertical and horizontal? If we did, quite probably, as Eddington remarks, we should come to an untimely end, and break our necks.

Obviously if we *visualize* our *plenum*, as made up of lines or particles, by necessity we visualize structurally a four-dimensional manifold. It should be noticed that a four-dimensional 'absolute void', or 'absolute nothingness', besides being non-sense, cannot be visualized at all, because it could have *no structure.*

We see that all metaphysical 'fourth dimensions' are not only non-sense, but usually indicate a pathological semantic disturbance. The intensity of such disturbances is often high, because it is entirely impossible for a sane person to deal with such meaningless noises. The victim is obsessed with attempts to do the impossible,—a semantically hopeless and painful task.

Such objectifications of terms are very dangerous and science should try, by proper emphasis, to eliminate them. Outside of science the term 'dimension' has *no meaning* and ought to be definitely abandoned in our speculations, for the sake of sanity.

The notion of 'time' as a 'fourth dimension' is by no means new. It appeared in a vague form centuries ago. The notion however was not formulated properly, and therefore was unworkable. Instead of helping science, it only hindered it.

Inspired by Einstein's work, the mathematician Minkowski, whose work had been mainly in the theory of numbers, began to work at the theory of manifolds of any number of dimensions. In 1908 he delivered his famous and semantically epoch-making address on *Space and Time* which fused geometry and physics structurally. In this address he insisted that the connection between 'space' and 'time' as given by the Lorentz-Einstein formulae is not accidental but exhibits that inner connection or structure to which we had not paid enough attention.[2]

In our *experience*, 'space' and 'time' can never be entirely separated, as already explained, and so Minkowski combined them into a higher entity which is called the 'Minkowski world'. In the world of experience the datum appears to be, not a place and a point of 'time', but the *event* or the *world point*—that is, a place at a definite *date.*

The graphic picture of a moving point is a *world-line.* Rectilinear uniform motion corresponds then to a straight world-line; accelerated motion, to one that is curved.

The *event* is the most elementary notion. We shall use it from now on in this work in the sense of a four-dimensional volume of space-time which is small in all four dimensions. We do not posit whether events themselves have structure or not, but it is preferable to assume that they have no space-time structure, which means that the event has no parts which are external to each other in space-time. The order of events is fourfold, as previously shown.

The aggregate or manifold of all point-events is then called the world. The point-events are given by four numbers representing the co-ordinates, three giving the 'space' co-ordinates, and the fourth the 'time' co-ordinate.

The term 'space-time continuum' or 'space-time manifold' is used often and implies that the numbers x, y, z, t, are to vary continuously.

In such a space-time continuum all happenings are structurally the intersections of world-lines, and if we could describe the world-lines of all points of the universe we would have a full account of the universe, 'past' and 'future'. We see that all physics, with the *rest of our problems*, must then be considered as a chapter of the *general* structural and semantic study of continuous manifolds of four dimensions.

But we are already acquainted with such theories. For instance, the internal theory of surfaces may be considered as a part of the subject in two and three dimensions. We have seen that different surfaces are characterized by the expression for the line element $ds^2 = g_{11}dx_1^2 + 2g_{12}dx_1dx_2 + g_{22}dx_2^2$, or by that group of transformations which leaves the line element invariant. We know already that in the E as well as riemannian geometries we have similar expressions and characteristic transformations.

If physics is to be considered a branch of the theory of four-dimensional manifolds, we should naturally look for some such transformations. The manifold represents the world, the *generalized** theory of relativity gives the desired answer. Minkowski proposed a postulate, which he calls the *postulate of an absolute world*, or the world-postulate which asserts the *invariance of all the laws* of nature in relation to linear transformations, for which the function $x^2+y^2+z^2-c^2t^2$ is invariant.

The reader is already familiar with the expression x^2+y^2, which gives the invariant length in E geometry in two dimensions, and $x^2+y^2+z^2$ which gives it in three dimensions. It would be natural to expect that in four dimensions we should have an expression of the type $x^2+y^2+z^2+t^2$ but in this case our expression is $x^2+y^2+z^2-c^2t^2$. It should be noticed that the above different types of expressions have different origins. The first two arise in pure geometry, and the last has its roots in physics. The problem was to bring an experimental expression into harmony with a familiar geometrical expression. Minkowski introduced the expression $ict = u_4$, where i is as usual the square root of minus one, $(i = \sqrt{-1})$. Then of course $-c^2t^2$ becomes $(ict)^2 = u_4^2$.

*I use the term 'generalized' to embrace the unified field theory and eventually the quantum theory, although, for our purpose, I utilize only the special and general theory.

If we change the lettering, and denote $x=u_1$; $y=u_2$; $z=u_3$ then our expression $x^2+y^2+z^2-c^2t^2$ becomes ${u_1}^2+{u_2}^2+{u_3}^2+{u_4}^2$, a simple formula for distance in four-dimensional geometry. It would not be profitable for us to speculate upon this substitution; it was introduced merely for the sake of mathematical, verbal treatment and can be easily translated back into the usual terms of c and t.

We have already seen that the expression $x^2+y^2+z^2-c^2t^2=x'^2+y'^2+z'^2-c^2t'^2$ is invariant under the Lorentz-Einstein transformation. This fact of invariance is fundamental, and it is well to convince ourselves that it is so. The Lorentz-Einstein transformation was $x'=\beta(x-vt)$, $y'=y$, $z'=z$, $t'=\beta(t-vx/c)^2$ where $\beta=1/\sqrt{1-v^2/c^2}$. As the co-ordinates y and z are equal in both systems, we can disregard them, and verify that $c^2t'^2-x'^2=c^2t^2-x^2$. Let us substitute for t' and x' the values given by the Lorentz-Einstein transformation. We have then:

$$c^2t'^2-x'^2=c^2\beta^2(t-vx/c^2)^2-\beta^2(x-vt)^2=c^2t^2\beta^2+\frac{c^2\beta^2v^2x^2}{c^4}-\frac{2c^2\beta^2tvx}{c^2}-\beta^2x^2-\beta^2v^2t^2$$

$$+2\beta^2xvt=c^2t^2\beta^2(1-v^2/c^2)-x^2\beta^2(1-v^2/c^2)=c^2t^2\frac{1-v^2/c^2}{1-v^2/c^2}-x^2\frac{1-v^2/c^2}{1-v^2/c^2}=c^2t^2-x^2;$$

since $\beta^2=\dfrac{1}{1-v^2/c^2}$. Similarly it is easy to show, if we take an event-particle, as, for instance, a momentary spark, which has the co-ordinates x_1, y_1, z_1, t_1, in one system of co-ordinates, say S, and let another event-particle occur in that system at x_2, y_2, z_2, t_2, that the formulae remain invariant. If we designate the distance between the two events by r, its value would be given by $r^2=(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2$.

In a different system, S', moving uniformly relatively to S, r'^2 would in general not be equal to r^2, but the expression $r^2-c^2(t_2-t_1)^2$ would be equal to $r'^2-c^2(t_2'-t_1')^2$, $r^2-c^2(t_2-t_1)^2=r'^2-c^2(t_2'-t_1')^2$.

The above expression is called the *interval* and expresses a most fundamental structural characteristic; namely, that the interval is invariant for all systems in uniform relative motion. This result is quite general and independent of the relative orientation of the axes, or of the angle the velocity has to the axes. The interval plays in the theory of Einstein a similar role which the pythagorean rule played in the E geometry.

Because of the finite velocity of our measuring signals, our formulae must involve finite velocity. Therefore the interval is the only actual measurement which we can ever make in practice. Hence its fundamental semantic importance.

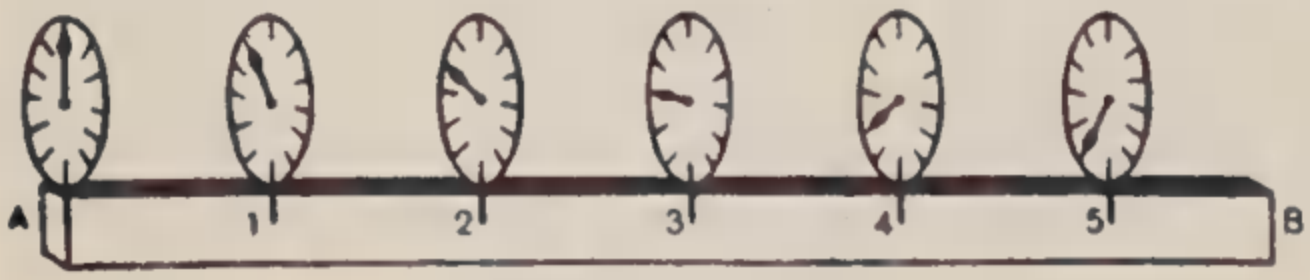

FIG. 1

Eddington gives a very fine diagram in explaining how intervals are measured. I reproduce it herewith (Fig. 1).

The expression ct, where c is the velocity of light, 300,000 kilometres per second, gives us the *distance* travelled by light in the 'time' t. It is natural to regard the velocity of light, which is a constant and translates easily into the language of length, as a unit of 'time'. In the Minkowski world it is customary, because of its convenience, to regard 1 second as the equivalent of 300,000 kilometres and measure lengths or 'times' in seconds or kilometres indiscriminately.

Let us imagine a scale graduated in kilometres, and clocks whose faces are also graduated in kilometres (1/300,000 of a second). If the clocks are set correctly and we look at them from A the sum of the reading of any clock and the scale division beside it is one for all because the scale reading gives the correction for the 'time' taken by light, travelling with unit velocity, to reach A.

If we lay the scale in line with the two events and note the clock and scale readings, t_1 and x_1, of the first event, and the corresponding readings, t_2 and x_2, of the second event, then $s^2 = (t_2 - t_1)^2 - (x_2 - x_1)^2$ where s represents the 'interval' mentioned above.

If we set the scale moving in the direction AB then the divisions would have advanced to meet the second event and the difference $(x_2 - x_1)$ would be smaller. But this is *compensated*, because $(t_2 - t_1)$ also becomes altered. When A is advancing to meet the light coming from any of the clocks on the scale the light arrives too quickly, and the reading of the clock appears smaller.

The net result is, roughly, that it does not matter what uniform motion is given to the scale, the final results for the interval s are always equal.[3]

We can now understand the vital importance of the minus sign with the 'time' co-ordinate. In fact, if in our equations all the signs were plus, using the 'space' and 'time' of one observer, one value of s would be obtained; but using the 'space' and 'time' of another observer, a different value would be obtained. With the minus sign for the 'time' co-ordinate, we see that we can have values of s which are equal for all observers. If the distances increase, the 'time' element increases also, and so the *difference* may not be changed, but with the positive sign this would not be the case.

We see that the interval s represents something which concerns only the events under consideration. The corresponding entity in ordinary geometry is *distance*, which is independent of the accidental choice of co-ordinates. The minus sign makes the geometry of space-time *non-euclidean*.

To familiarize ourselves with what has been already explained about simultaneity and the geometry of space-time, we will work it out once more, but now by the Minkowski method.

It will be enough to use two dimensions, one represented on the X axis, the other on the T axis. Let us consider three points A, B, C, at rest in our system O on the X axis (Fig. 2). In our space-time they will be represented by three parallels to the T axis. Let C be midway between A and B so that $AC = CB$. Let us assume that light signals are sent in both directions from C at the moment $t = 0$. We assume that the system is 'at rest', which means that

the light signals propagate themselves to the right and to the left with equal velocities. Hence we can represent them by straight lines equally inclined to the X axis. These lines are called 'light-lines'.

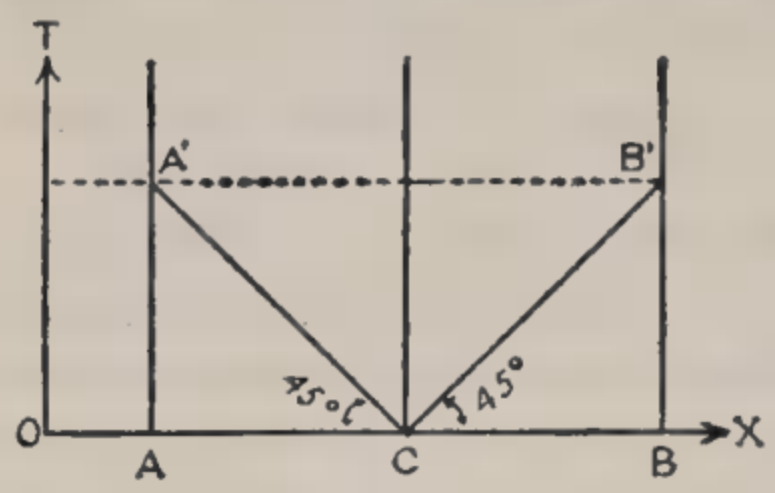

FIG. 2

The points A', B' which are the intersection of the 'world-lines' of the points A and B with the light-lines give us the 'times' at which the signals arrive. It follows from the drawing that $A'B'$ is parallel to the X axis, which means that A' and B' are 'simultaneous' (equal 'times').

Let us now take another case in which our points A, B, C, move uniformly with an equal velocity (Fig. 3). Their world-lines will also be parallel to each other but *inclined* to the axis. In the drawing the light-lines will be represented by similar lines but their intersections with the world-lines of A and B will not be on a parallel to the X axis, and so they will not be simultaneous.

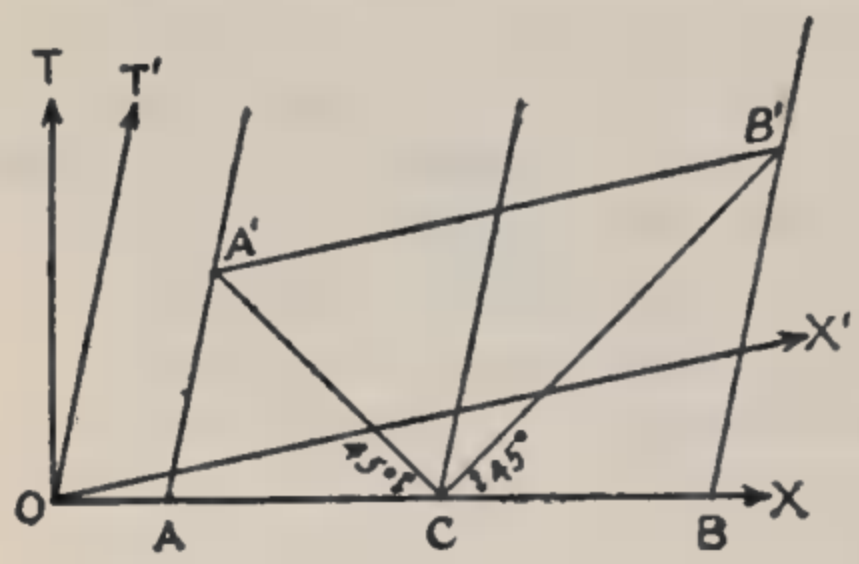

FIG. 3

We should notice that an observer who moved with the system in the direction OX' would be perfectly entitled to claim that A' and B' are *simultaneous to him*. His co-ordinate system would be $OX'T'$, in which the points A' and B' are on a parallel to *his* X' *axis* as he is *at rest* in his system $OX'T'$. The world-lines A, B, C, are parallel to the T' axis because the points are supposed to be at rest in this system and hence the x's have equal values for all t's.

An important point should be noticed; namely, that we have only one space-time and that the indefinitely numerous ways different observers partition their 'space' and 'time' represent merely the indefinitely many ways in which it can be partitioned. If we keep the whole of it under consideration we see that we cannot divide it into 'space' *and* 'time', as any subdivision has both aspects.[4]

The Minkowski method of representation makes the change in our measurements of length, as given by the Lorentz-Einstein transformation, very obvious.

A measuring rod is not purely a 'spatial' configuration, as in the actual world such a thing does not exist, but it is a space-time configuration.

Every point of the rod exists at each moment of 'time'. We see that in space-time we cannot represent our rod as a segment on the X axis but must represent it structurally as a *strip* in the XT plane. We assume here for simplicity that the rod is one-dimensional (Fig. 4).

A rod which is at rest in a system is represented by a strip parallel to the T axis. If it is moving, its strip is inclined to the T axis. The 'contraction' does

not affect the strip at all but it is rather a section cut out of the X axis. In actual experience, it is only the strip as a manifold of world points which has *physical reality*, and not the cross sections, which, as we see, are not equal on different axes. The 'contraction' is not a change in 'physical reality' but merely a consequence of our way of regarding things. We see that the notorious argument as to whether the 'contraction' is 'real' or 'apparent' is based on a misunderstanding. Born gives an excellent example. If we slice a cucumber in different directions it is fallacious to argue that the smallest slice which is perpendicular to the axis is the 'real' one and the larger oblique slices only 'apparent'. Similarly in the Einstein theory, a rod has various lengths according to the motion of the observer. One of these lengths, the static length, is the greatest, but it is no more 'real than any other. Similar remarks can be made about 'time'.

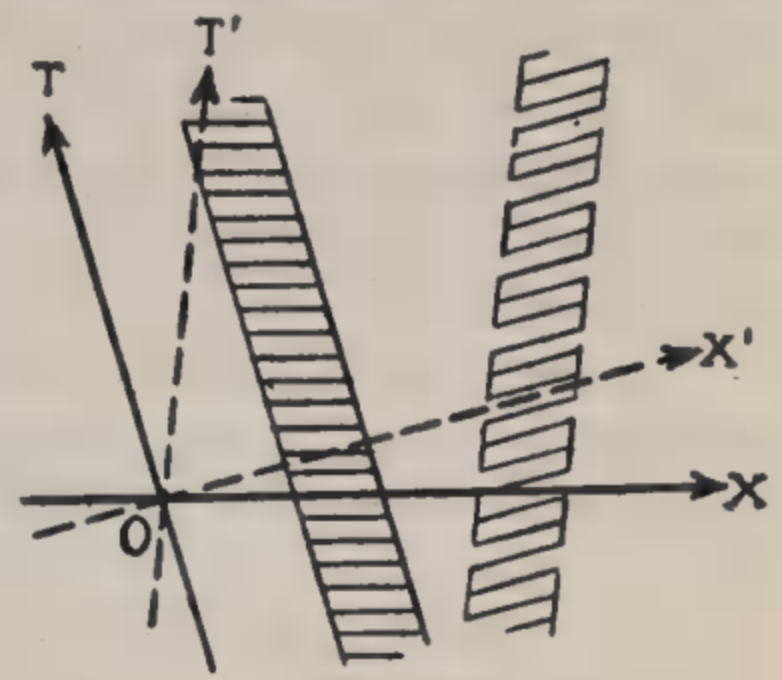

FIG. 4

Attention should be given to one extremely important semantic point concerning the Minkowski four-dimensional world. We already know that for our nervous systems the passing from dynamic to static, and vice versa, is a most vital structural problem. The first step of this translation has already been given in the notion of the 'variable'. The calculus carried it a step further. In the Minkowski world we reach the complete solution of the problem.

As Keyser points out in his *Mathematical Philosophy*, we had two verbal methods of dealing with 'time'. One was the method of Newton, the method of the structural importation of 'time'. From the objective dynamic world of the lower order abstractions, 'time' is imported into the static world of the higher order abstractions. We import it with 'motion', we say things 'move'. Such language is structurally unsatisfactory, even on the earlier level of our development. It hampers analysis, and is contrary to the structure and function of the *human* nervous system. It breeds tremendous metaphysical impasses, and is ultimately based on semantic disturbances due to identification.

If we introduce dynamic, shifting entities into static higher order abstractions, rationality is impossible and we drift toward mysticism.

A very real semantic problem appears here. We want to give the best possible account of the structurally *dynamic* world around us; yet our higher order abstractions are structurally *static*, and for their proper working they must use static means. Here seemed to be an impasse which for milleniums had defied solution. 'Philosophers' of different schools were preaching and teaching that we should never be able to be 'rational' and understand this world and ourselves. Anti-intellectual schools began to flourish, to the bewilderment of all.

The issue, after all, was simple, the moment some one discovered and stated it. We did not need to change either the world around us or ourselves;

we simply had to discover a structurally new method of dealing with the old problems without changing them.

The new method is given by Einstein and Minkowski. Instead of making the static world of higher order abstractions dynamic, which cannot be done at all without producing semantic disturbances, they invented structural methods for dealing with the dynamic world by static means. The key was found in the handling of the troublesome factor, 'time'. Minkowski decided to put 'time' in its proper place by introducing the structurally new four-dimensional world.

In the case of particle P, we habitually used to say that the point P at an instant t was at a 'space' point (x, y, z). At the instant t' it was at the point (x', y', z') , . We always needed four numbers, which gave us the where and when in respect to some frame of reference.

As we have seen, Minkowski decided to look structurally at this tetrad of four numbers (x, y, z, t) 'as-a-whole'. In other words, he placed himself on a higher level of abstraction. He took under consideration the older results, combined them, and called the combination by one single name, the 'world point'. Such a world point has also four numbers (not 3 plus 1, but just 4). A world made up of such points is a four-dimensional world in which all the points co-exist. The flux of the lower order abstractions and 'time' is abolished. There is no more 'motion' in a 'flow of time'. In such a world the term 'where' has completer structural meanings, it has absorbed the *when*. If we ask, where in such a world the particle P is, we answer, at the point (x, y, z, t). Where is the particle P'? At the point (x', y', z', t').

We see that the particles of such a world *are never 'the same'*; they do not 'change' or 'pass'; they co-exist, and all is *static*. In this way the three-dimensional dynamics become four-dimensional statics.

It should be noticed that we are now dealing with a language of new structure, uniquely befitting the structure and function of our nervous system. Of course we have altered nothing in the world around us. The example of the moving picture and the static film as given before is an excellent example of this structural innovation.

The fact that in this new world nothing repeats itself because it has a different date, unless the time-lines are closed, has very far-reaching consequences, of which we have already spoken and which we will analyse in more detail later on.

CHAPTER XXXIX

GENERAL REFLECTIONS ON THE EINSTEIN THEORY

> For, beyond the bounds of science, too, objective and relative reflection is a gain, a release from prejudice, a liberation of the spirit from standards whose claim to absolute validity melts away before the critical judgment of the relativist. (45)
>
> MAX BORN

It is extremely instructive to follow the elements of identification as they appear in the evolution of the Einstein theory.

We know that the results of the Michelson experiment which disclosed that light is propagated in all directions with equal velocities for all observers, irrespective of their relative velocities, could not be reconciled with the older mechanics. These results disturbed the physicists profoundly and attempts were made to solve this apparent impasse. In what follows we shall have to analyse incidentally the activities of some of our great scientists, men who have added enormously to our knowledge, and this fact should be appreciated. What we have to say is not intended as criticism—far from it—but simply as a structural and a semantic analysis.

The feeling that we objectify unduly and that we should not use a language of 'is' of identity, but that we should use an actional, behaviouristic, functional, operational language and methods, is not new in science, although the need was not formulated structurally, it is true, and therefore it never became a workable foundation. The main successes in these fields were rather accidental, and were the personal prerogatives of those few men whose psycho-logical make-up urged them to achieve. Objectification, which as we know, is a semantic ascribing of *objective* existence and values to terms, was bound to make its appearance somewhere.

This struggle against identification is apparent in all science, but it will suffice to point out the most striking example in the relation of the works of Lorentz and Einstein. Lorentz objectified, Einstein did not. We here come across a tremendous semantic fact which has to do with the *interpretation* of mathematical formulae. Lorentz on elaborate and difficult grounds, connected with Maxwell's field equations, produced what is usually called the Lorentz transformation. He gave it an *objectified* interpretation. Einstein introduced an entirely different fundamental interpretation of the structural *principle* involved. The formulae look alike but they now have different and very simple meanings.

Hertz, whose epoch-making discoveries made wireless possible, advocated long ago what is termed the phenomenological point of view, which in our language is approximately the actional, behaviouristic, operational, functional language and method. In his writings he implicitly refused to use the vicious

term 'is' of 'identity', and so to objectify his terms, which refusal in picturesque language he expressed as a refusal to legislate about 'essences'.

The old E and N language of 'absolute space' and 'absolute emptiness' were for a long while structurally unsatisfactory. Physicists felt that somehow they could not deal with it, but it never occurred to them that this 'absolute nothingness' is objectively meaningless, and that therefore no one can possibly deal with it. Not knowing that, they politely called this non-sense a 'meta-physical question' and evaded issues by leaving the solution in the hands of 'philosophers', never to be solved.

By now I hope that the reader is quite aware that meaningless problems cannot be solved by any one, and that there are no such things as *meta*-physical questions. There may, however, be a question about *enlarging* the domain of physics.

Being forced to abandon this 'absolute emptiness', physicists went to the other extreme and postulated some kind of 'material' ether. Let us note that such a postulate involves structurally the 'is' of identity and objectification. Lorentz in opposition to Hertz postulated an 'ether' which was 'motionless' in 'absolute space'. Note that here we have a perfect example of structural *objectification of terms*. 'Absolute space' is for him semantically some kind of 'absolute emptiness', which, not being satisfactory for the physicist, is filled with some 'material', 'motionless' ether. 'Motionless' is itself an objectification of language, as such a term has here *no* physical or objective meanings at all.

In pursuing the speculations on *objectified terms* (semantic disturbances) it was natural to expect, as the earth is not at rest with respect to the sun, the other planets . , that some 'ether wind' or 'ether drift' should appear which would make the constant velocity of light impossible for observers moving with different relative velocities. But these structural expectations were not fulfilled. The velocity of light, as shown by many experiments, was a constant for all observers. The 'motionless material ether' also became structurally impossible, as might be expected, if we stop objectifying terms.

In 1892 FitzGerald suggested an *objectified* theory, assuming 'absolute' 'length' and 'time' *superior to measurement*, which involve identification and do not allow the use of the actional, behaviouristic, operational, functional attitudes, language, and methods. FitzGerald assumed that every body 'moving' with the velocity v in the 'ether' is shortened in the direction of motion. It should be noted that every mention of 'shortening' or 'contraction', *presupposes some 'absolute'* standards of 'rest' or 'motion' or 'length', which do not, and cannot, exist outside of our skin, but are only semantic disturbances, inside our skin, which occur when we identify and ascribe *objective* existence and value to *terms*.

How deeply and completely these objectifications permeate our daily and scientific lives is best shown again in the case of Lorentz. Even in 1917, in his Haarlem lectures, he expressed structural hopes that a 'material', 'substantial' ether can be preserved, that 'space' and 'time' can be sharply separated, and that 'simultaneity' can have an absolute meaning.

In the *Theory of Relativity* of Whitehead, and in some others writers who deal with the theory of Einstein, and particularly in all critics of Einstein, we find a similar objectification of terms.

They still *feel* the older E and N 'absolute emptiness', 'absolute space', 'absolute time', to which *terms* they ascribe structural objectivity. In such works the term 'contraction' is used frequently.

Let me recall the mechanism of objectification. If we do not reject explicitly and implicitly the 'is' of identity, we automatically identify different orders of abstractions and ascribe objective characteristics to terms. Thus the term 'time' which represents a label for a feeling inside our skin, is given an objective evaluation. If 'objective' it must have a 'property' of 'simultaneity', a semantic process taken over from comparing two objective sticks when the two ends are made to coincide. On the objective external level, we never deal with 'time' but we simply *compare processes*. When we select an arbitrary unit-process on the objective level, whatever we might *say* that it 'is', well, it *is not*, and the difficulty is found exclusively in the use of the 'is' of identity.

If we abandon entirely the 'is' of identity, we stop objectification, we do not ascribe objective existence and values *outside* our skin, to terms and semantic reactions *inside* our skin. But then of course we have to change the *structure* of our language; as otherwise the old *s.r* will continue to play tricks on us. An actional, operational, functional language of *order* is the structural solution of our semantic difficulty.

If we objectify 'space' into 'absolute space', we must objectify it as 'absolute emptiness' for only such an 'absolute space' can be at 'absolute rest', that is, static in the E or N sense. Similarly only objectified 'time' can have the 'property' of 'absolute simultaneity'.

If we realize that these 'absolutes' are only the semantic objectifications of terms, (where the activities of the lower nerve centres are structurally ascribed to the activities of the higher nerve centres and vice versa), we begin to differentiate between different order abstractions, and to keep them differentiated. In terms of our structurally new language we become 'conscious of abstracting', and then habitually *and unconsciously* use the behaviouristic language and methods of *order*.

If we picture this 'absolute emptiness' or 'absolute nothingness' (which cannot be done successfully, as it has no meaning), and try to compare it with a plenum, or 'fulness' (a cloud of smoke, for instance), we see at once that only this 'absolute emptiness' can be static, homogeneous . , a condition that is impossible with a dynamic fulness.

Perhaps we can now appreciate the tremendous semantic significance of the Einstein theory, which introduces structurally a *non-objectified*, human, sane attitude of proper evaluation toward this world. We should not be surprised to find that a $\bar{A}$-system which is an inevitable general structural concomitant of the $\bar{E}$ and $\bar{N}$ systems of geometry and physics should *formulate as a general structural and semantic issue* what the $\bar{E}$ and $\bar{N}$ systems have *done* in their special fields, without such general formulation.

From our structural point of view there is no retreat; the Einstein work is irreversible. In the younger scientists of today the non-objectified attitude toward *terms* of 'space' and 'time' is already an accomplished semantic fact, entirely independent of what future experiments may show. For experiments can never justify identification, and so can have no detrimental effect upon this fundamental and most beneficial structural, linguistic, and semantic revolution. Our $\bar{A}$ task was to formulate these issues *in general* so as to make us conscious of them; and I assume that it is at this semantic point that the tremendous value of Einstein's work will manifest itself in life. Indeed we shall see later on in this volume that the newer quantum mechanics, which have begun to spring up rather rapidly, is made possible only by the semantic background imparted *unconsciously* (as yet) to younger physicists by the Einstein theory. It is my hope that the present work may make the above issues *conscious*, and so enable us not only to impart this semantic attitude more easily and with less labour but also to benefit by them more universally in our *daily life*. The problems of science and life do not differ in this respect. In both we are equally hampered by semantic disturbances, 'emotional stupors', identification, and similar difficulties, the elimination of which means better adjustment for all of us, as well as swifter progress in science.

A study of the history of science shows how slow and painful scientific progress has been. Now we begin to see why. 'Geniuses', as history shows, are men who at least in some fields are freer from identification and false evaluation than others. They are not hampered to a similar extent by 'emotional stupor'; hence they can evaluate the old *anew*. Lorentz, for instance, produced the formulae, but his objectifications *prevented* him from evaluating properly the new formulae. As a fact of history the formulae of Lorentz were discovered by Voigt a number of years before, but identification made impossible the evaluation of these formulae, and so *delayed* the discovery of the Einstein theory. This factor of identification can be found all through recorded history as a retarding semantic blockage.

If we could find methods of eliminating these semantic disturbances, an extremely hampering, paralysing psycho-logical factor would be eliminated, and 'geniuses' could be made the rule rather than the exception. Let me say again: in the old days morons were made and geniuses were born; in the new days, perhaps, this can be reversed, and *morons will be born but geniuses made.* We witness something of this kind among the younger post-einsteinian physicists, where the number of 'geniuses' is growing rapidly, in spite of the fact that the above structural issues are not as yet consciously applied in general education. The secret of creative work is freedom from structural bondage, and particularly the structural semantic bondage of words.

The reader should not assume that the few simple structural explanations given in this book exhaust the Einstein theory. I have not even attempted to summarize the theory; I have only given a few semantic facts, which belong to general semantics and to the theory of knowledge. The Einstein theory is

indeed such a tremendous structural linguistic achievement that quite probably its full semantic significance and meanings will not be worked out for many years to come. We have given here only the minimum of explanation necessary for our special purpose.

The historical development of a theory has usually little to do with the semantic importance of the theory or its deeper meanings. The constancy of the velocity of light for all observers, which started the ball rolling, was an historical beginning and it served its purpose well, though the objectified 'contractions' and formulae of FitzGerald and Lorentz also did their share, as they helped Einstein and Minkowski to produce their epoch-making structural challenge to old prejudices such as 'absolute space' and 'absolute time', which were semantic remains of a primitive, perhaps pre-human, remote past. Once this is accomplished, no matter how, there is no return possible. Of physical structural facts, all that we need is the *finite* velocity of the propagation of events,* which as we already know involves far-reaching structural and semantic issues. Of the psycho-logical issues involved, we need only to eliminate semantic disturbances which still occur when we copy animals in our nervous processes and do not discriminate between different orders of abstractions—which animals do not recognize. This elimination can be done by training in the $\bar{A}$ methods explained before, with the net result that we become 'conscious of abstracting' on different levels and so can instinctively and by feeling discriminate habitually between orders of abstractions, which structurally and semantically could not be done by the old disciplines.

The theory of Einstein has manifold applications but we need only mention a few, which we shall utilize later on.

First, and above all, there are no possible 'absolute' meanings to 'space' and 'time', beyond the relations established by measurements. The structure of our language involving 'space' and 'time' should be similar to the structure of experimental facts, which ultimately show the impossibility of sharply dividing them.

If any one challenges this statement, he could not *a priori* be criticized. Such criticism would be entirely against the whole tendency of the present work. But such a person might be approached with no little curiosity and expectation. He could be asked: 'You claim that you can absolutely divide

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*'But,' some reader may ask, 'though you assume a finite velocity of propagation, may it not happen that some day an "infinite" velocity will be discovered?'

Such a question would show that the reader has missed the point in the present work. We are confident in saying that an 'infinite' velocity has no meaning, and that no matter what we discover, this will never be discovered. This becomes still clearer if we use the differential *definition* of 'velocity'. Velocity is defined as the 'time' derivative of 'space' travelled. If 'time' is taken as zero, or if we have 'no time', there can be no 'time derivative' by our very assumption, and, therefore, no 'velocity'. There is, therefore, no danger that we shall ever discover in the actual world an 'infinite' velocity.

"space" and "time" on the objective level. That would be an epoch-making structural discovery. *Please demonstrate how to do it.*'

The fact is, of course, that he cannot demonstrate the process, because he refers to identifications inside his skin; yet he is claiming to be able to show it *objectively* outside his skin. That ends this problem.

While speaking of Einstein's Theory, it will be well to mention a few of the many structural differences between the older newtonian and the new einsteinian mechanics.

In the N-system, relative velocities were simply added $W_N = v + v'$. In the einsteinian system which we will denote by $\bar{N}$, it is not so structurally simple. We must introduce the finite velocity of propagation of our signals, which *alone* give us the data, and so

$$W_{\bar{N}} = \frac{v + v'}{1 + vv'/c^2}$$

The above formula involves the remarkable constant, c, the velocity of light. If we assume in the above formula that our velocity v' is equal to the velocity of light, c, we would have

$$W_{\bar{N}} = \frac{v + c}{1 + vc/c^2} = \frac{v + c}{1 + v/c} = c.$$

This means that the addition of some velocity to the velocity of light does not alter the velocity of light, which thus appears as a *limiting* velocity.

This applies to the difference of velocities where

$$W_{\bar{N}} = \frac{v - v'}{1 - vv'/c^2}$$

Let us here give an example of Eddington's. Let us assume two relative velocities each differing by only 1 km./sec. from the velocity of light. Let us say that one is 299,999 km./sec. and the other 300,001 km./sec. Now let us calculate the relative velocity. This relative velocity will be found to be 180,000,000,000 km./sec. For in our formula $v - v' = (c+1) - (c-1) = 2$, and

$$(1 - vv'/c^2) = 1 - \frac{(c+1)\ (c-1)}{c^2} = 1 - \frac{c^2 - 1}{c^2} = 1 - 1 + 1/c^2 = 1/c^2,$$

whence $W_{\bar{N}} = \dfrac{2}{1/c^2} = 2c^2 = 2 \times 300{,}000 \times 300{,}000 = 180{,}000{,}000{,}000.$

We see that a particle which might try to overtake light by having a velocity of one km./sec. greater than the velocity of light could never succeed. When the velocity 299,000, for example, was reached, the particle would find itself further away from its goal than when it started.[1]

Similar general considerations apply to mass. If we designate the mass of a particle at rest by m_0 its mass in motion

$$m_{\bar{N}} = \frac{m_0}{1 - v^2/c^2}$$

As the denominator is smaller than unity the mass in motion, $m_{\bar{N}}$ is larger than m_0, the mass at rest. In the limiting case, when the velocity would become equal to c the denominator would become zero and our mass $m_{\bar{N}}$ would tend

toward infinite values, which is another way of saying that it is physically impossible.

In the N-system we had two kinds of energy; one was called *vis viva*, or kinetic energy, and was represented by $T=mv^2/2$; the other was called potential energy, or capacity for work, and was denoted by U. The law of conservation of energy in the N-system was expressed by the statement that the sum, $T+U=E$, or the total mechanical energy of a system remains constant (zero variation) during the motion of the body.

We see that as the above formula involves the terms m and v, the older formulae for energy must be altered, especially since they do not survive a Lorentz-Einstein transformation. It is found that

$$T_{\bar{N}}=c^2\,(m-m_0),\text{ or } m=m_0+T_{\bar{N}}/c^2,$$

which formula appears rigorous as a definition of kinetic energy even if members of order higher than the second are taken into account. In words, the mass in motion differs from mass at rest by the kinetic energy divided by the square of the velocity of light.

This expression suggests immediately that the static mass, m, is similarly related to the energy content in the body at rest. Generalizing our results we would have $m=E/c^2$, an equation which holds generally between mass and energy. This fact has been called by Einstein the law of the *inertia of energy*. It has been verified repeatedly by experiments, and is one of the most striking structural results of Einstein's theory. The above statement means that the two fundamental notions of 'mass' and 'energy' are equivalent and thus we have a clearer vision of the structure of 'matter'. The two older structural laws of 'conservation of matter' and of 'conservation of energy' become fused into one. Mass becomes structurally and verbally nothing else than energy concentrated at a point, and it appears as a form of energy manifestation.[2]

The above considerations have also led to a revision of our structural notions about 'energy' which we do not need to explain here. Suffice it to say that the old 'potential' energy is not associated structurally any longer with any features of this world. It can be made to vanish by a proper selection of co-ordinates, hence it is no longer considered as energy of any kind.[3]

With the Minkowski world we became acquainted with a new *language* which represents structurally more nearly the facts of experience (lower order abstractions) and shares the structure of our higher order of abstractions. So we have the language of 'space-time'. How about 'matter'? The bumping against something hard is not to be disregarded. True, *we need a language of new structure, but that is all.* In the Einstein theory, 'matter' of course is not treated separately as such. It is an offspring of the field, and is connected with the curvature of the world. The reader should not be surprised to find that the Minkowski world, which has accelerations, must be curved in this structurally new form of representation.

We have already defined a most fundamental entity called 'action'. Naturally in a space-time manifold, energy multiplied by 'time' should be a more fundamental entity than energy, and we call it 'action'. When we speak about

some continuous material present in 'space' and 'time' we speak in terms of density. Density multiplied by a three-dimensional volume of 'space' gives us mass, or what appears as its equivalent—*energy*. From a four-dimensional, or space-time point of view, density multiplied by a four-dimensional volume of space-time gives us action. We see that the multiplication of density by the three dimensions of 'space' gives us mass or energy. A fourth multiplication by the dimension of 'time' gives us mass or energy multiplied by 'time' which becomes action by definition. It is obvious that, structurally, action must be more fundamental than the older quantities.

In terms of curvature, action represents the curvature of the world, because where we find 'action', we also find 'matter', acceleration, gravitation , .[4]

'Action' is fundamental, because structurally in a four-dimensional metrical manifold it takes the form of the simplest integral *invariant* that can exist at all. On this form of action Maxwell's electromagnetic theory is built. The quantity action appears as a pure number,[5] a unique, specific relation which conditions structure.

We should expect that the action represented by the number 1 would be most interesting and would eventually represent the indivisible atom of action. The modern quantum theory seems to favour such a point of view.

When we encounter a pure number having such crucial significance in this world we should not wonder that such a number intrigues us. As yet it is impossible to state that action cannot have fractional numbers. What, then, would the action represent?

Eddington suggests that the number may represent a *probability* or some function of a probability.

We combine probabilities by multiplication, but we combine actions in two regions by addition. We see, therefore, that the logarithm of a probability gives the function indicated and Eddington suggests the provisional equivalence of action with the negative of the logarithm of the statistical probability of the state of the world around us. Such a suggestion is extremely appealing and important because the principle of *Least Action* can be stated as the principle of *greatest probability*. The laws of nature appear to be such that the actual state of the world is represented by that which is statistically the most probable![6]

That such structural conclusion can be drawn at all is of tremendous semantic importance for us because, as we are *abstracting* in different orders all through, the only appropriate language in which we can eventually hope to speak correctly, is the language of probabilities, statistical averages , .

Action is one of the terms of pre-einsteinian physics which has survived unmodified, the only other one being entropy. The law of gravitation, the laws of mechanics, and the laws of electromagnetism, can all be, not only summed up, but also deduced, from a single principle of least action. This important structural unification was accomplished even before the advent of the einsteinian theory, and only the addition of gravitation to this list is new.[7]

In this brief structural and semantic survey we have had neither the opportunity nor the necessity of analysing the general theory of Einstein, which embodies and unifies most of the laws of mechanics, that of gravitation included.* In this unification lies the unrivalled grandeur of the theory. As we shall see later, the newer quantum theories have been already very much influenced by the Einstein theory. As all possible theories are dependent on *human ingenuity* and never can be the events themselves, we can rest assured that once freed from 'emotional stupors' and semantic disturbances, the world will not be long in producing a whole structurally unified system of science.

In our discussions we deal with 'apparent', 'real', 'actual', and similar *m.o* terms. We should recall that mathematics is *exclusive* in one respect; namely, that it has no content. It is entirely a product of higher abstractions created by definition from undefined terms. We have seen that mathematics must be considered as a language of special structure which is, however, similar to the structure of the world around us.

Our daily *A* language, among others, being based on the 'is' of 'identity', can never give a structurally satisfactory picture of this world or ourselves, but actually prevents such an achievement. Having abandoned a language which leads to identification, we shall be able to apply a new language, with new structure, by which we achieve better means for representing the events around us. From this point of view, mathematics and our daily language do not differ. Terms, being *not* the things they represent, must by necessity be creatures of definitions and undefined terms. The solution of many baffling semantic problems is found in the *structure* of a language which involves different semantic and unconscious attitudes.

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*In fact, a few months ago, Einstein and Mayer succeeded in reducing the laws of gravitational and electromagnetic fields to a single basis. This was accomplished by the aid of a very revolutionary mathematical discovery that it is possible to introduce into a 'space' of *n* dimensions, vectors with *m* components. Although at present the results of the quantum theory are not included in this theory, there is no doubt that shortly, because of this mathematical discovery, these will be included in a *generalized* theory of relativity.

PART X

ON THE STRUCTURE OF 'MATTER'

Rather against my better judgment I will try to give a rough impression of the theory. It would probably be wiser to nail up over the door of the new quantum theory a notice, "Structural alterations in progress—No admittance except on business", and particularly to warn the doorkeeper to keep out prying philosophers. (149)

A. S. EDDINGTON

CHAPTER XL

THE OLDER 'MATTER'

> And yet when I hear to-day protests against the Bolshevism of modern science and regrets for the old-established order, I am inclined to think that Rutherford not Einstein, is the real villain of the piece. (149)
>
> A. S. EDDINGTON

> Micro-mechanics appears as a refinement of macro-mechanics, which is necessitated by the geometrical and mechanical smallness of the objects, and the transition is of the same nature as that from geometrical to physical optics. (466)
>
> E. SCHRÖDINGER

From the dawn of history, man has had to deal with different bits of materials, some hard and solid like stones, some soft like fruit or flesh, some liquid. In remote antiquity air and gases were not considered as 'matter'.

In those days 'matter' was structurally only what could be seen, or felt, or touched . : anything else was some kind of 'spirit', and everything 'existed' in an 'absolute void'. But even in remote antiquity our primitive ancestry could not miss the fact that the bits of materials they dealt with could be divided into smaller bits. Naturally, if we can subdivide bits into smaller bits, an interesting question arises: How far can this division be carried on? It seems that Democritus (about 460-360 B.C.) was the first man on record to formulate an atomistic theory. He already postulated structurally a subjective world picture, to be contrasted with an 'absolute' or objective world in which 'motion' was all important. This theory started us on the mechanistic road formulated for macroscopic events, and also on the road of individualization, the study of smaller and smaller bits of materials and the search for some unit bricks out of which this world appeared to be built; all of which was already a search for *m.o* structure.

With the advent of chemistry some further fundamental structural light was thrown on the problem of individualization. It was found that certain materials, as, for instance, iron, copper . , remain one material, no matter how far we carry our subdivision. These were called 'elements'. At present we recognize 92 elements, a number which is supposed to represent all possible elements. Out of these a remaining few were at first predicted theoretically, and just the other day discovered experimentally. All other materials do not stand division so well. At some stage they decompose into their elements. The smallest bit of one of these last materials which still has the characteristics as the bulk, is called a molecule. The molecule is found to be built up from atoms of the elements. For instance, the molecule of water still has the characteristics of water, and consists of two atoms of hydrogen and one atom of oxygen, which are no longer water but elements of entirely different characteristics.*

Electrochemistry taught us in the meanwhile an important structural lesson; namely, that definite electrical charges are combined with the atoms. Such electrified atoms are called 'ions' (Greek for traveller). For instance, a

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*The above statements are over-simplified, but satisfactory for my purpose.

molecule of water is broken up into a positively charged hydrogen ion consisting of two hydrogen atoms, and a negatively charged oxygen ion consisting of one oxygen atom.

But electricity had more structural surprises in store for us. About 1880 new facts were discovered. One of them was that a moving electrical charge has the effect of an electrical current; namely, it can deflect a magnet just as a current does. Such moving electrical charges were called convection currents, and the fact that they produce effects similar to an electric current led J. J. Thomson to a surprising conclusion. According to the Maxwell theory of electromagnetism a certain amount of energy must be associated with every electric or magnetic field. If an electrical charge in motion can produce magnetic effects, hence energy, it was concluded, and verified by experiment, that energy was required to set an electrical charge in motion. From which it follows structurally that an electrical charge possesses a characteristic in common with other materials; namely, inertia, which can be overcome only by the application of energy. This inertial mass of the electric charge was called electromagnetic mass.

Here we see two fundamental structural issues involved. One is that electricity seemingly has an inertial mass similar to that of 'matter'. The other is, that in convection currents, we find means to study electromechanical parallelism, and so discover the relationship between electrical and mechanical theories.

In the year 1895 Lorentz proposed the electron theory. He assumed structurally that moving molecules contain electrical charges and so produce convection currents. These charges are further assumed to be one electrical quantum and were called electrons. The electron theory proved to be enormously fruitful and all further advance in our structural knowledge is intimately connected with it.

As knowledge advanced, the more convincing became the structural evidence for some electronic theory. As already noted, a moving electrical charge produces magnetic effects. So also should a moving electron, whence we succeeded in accounting for magnetic effects in terms of moving electrons which in this case represent molecular convection currents.

Granting this, a revolving electron should represent also a small magnet and a mechanical gyroscope as well. Einstein in 1915 verified this assumption by an experiment. If these structural assumptions were true, then, by a quick reversal of magnetism, a soft iron rod should turn by a slight but definite amount. The reverse effect has also been verified; namely, a soft iron rod when rotated rapidly around its axis becomes magnetized.

The discovery of radioactive materials was also of enormous structural importance, as it gave us a means of studying directly the rays emitted from these materials. The rays were found to be of three kinds and were called by the first letters of the Greek alphabet. The α-rays have been found to be similar to positive rays, the β-rays similar to cathode rays, and finally the γ-rays to Röntgen-rays. Further investigation revealed that the α-rays were atoms

of helium charged with a double positive charge of electricity, and that the β-rays were negatively charged particles with the charge of an electron.

These few remarks already make apparent the structural fact that electromagnetic phenomena exhibit characteristics quite similar to those of 'matter', some of their processes are atomic, they have inertia , .

In the older days we tried to apply macroscopic mechanical structural laws to electromagnetic phenomena, but were not very successful. The laws which applied to those sub-microscopic levels were seemingly different from those which applied to gross macroscopic levels, just as the psycho-logics of the individual differ from the psycho-logics of the mob.

An epoch-making semantic step was taken by Rutherford when he formulated the electromagnetic theory of the structure of 'matter'. In this theory the atoms represent complex structures built up of positive and negative electrons, and their number and arrangements (structure) determine the chemical and physical characteristics of the atom in question. The old structural dogma of the immutability of the elements became untenable; and today, theoretically, and in a few instances experimentally, it has been established that a transmutation of elements is not only a possibility but a rather well-established structural fact of 1933.

It should be noticed that once more one of the fanciful linguistic, structural, 'infinities' has been abolished. The elements appear as transitory processes with a 'life' of a *limited* span of years. The experimental structural evidence which physicists and chemists have gathered is overwhelming and, though the *positive* theories (verbal structures) may not always be satisfactory, the *negative* results leading to the rejection of the older theories are conclusive. This point is of supreme structural importance to us.

A short description of the different atomic models comes later in this chapter, but first something must be said about the old quantum theory, which represents at present the central problem in science and out of the solution of which the most revolutionary consequences are bound to follow.

The main problems of the quantum theory may be described and contrasted somehow as follows. If we take a line $X'X$ and select a point O as the origin, we may fix the position of a point P on this line by the co-ordinate x. In practice we find the values of x by measurement. If we assume that x varies *continuously* we can expect that by refined measurements we can find values of x as close together as we please.

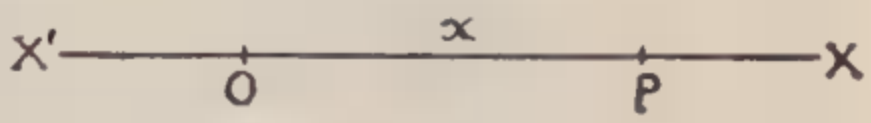

FIG. 1

Experiments show that for the processes going on 'inside of the atom' the structural conditions are somehow radically different. In comparing them with the above example, we should have to give our point only the freedom of occupying certain discrete points, let us say 1, 2, 3 . , but all fractional values, such as 1/2 or 2/3, would be impossible. If the only possible values of x are whole numbers, then the possibility of finding values of x as close together as we choose in order to make more precise measurements is excluded. If we find,

for instance, that our x is neither 1 nor 2 nor 4, and cannot be larger than 4, the only solution is that our x must be 3.

These conditions, which are actually found in the atomic mechanics, represent an entirely new and unexpected structural state of affairs. The whole of the older quantum mechanics can be summarized in the statement, that its peculiarity consists in the fact that structurally characteristic *discrete numbers* make their appearance, and that the processes 'inside the atom' are to be described by *discrete* numbers.

The usual classical quantum mechanics demands that x be allowed to take all possible continuous values, but that the *integral values* of x represent the so-called stationary states through the quantum conditions. Under such conditions the intermediate fractional values had no meanings.

It is safe to say that, when in 1900 Planck formulated his quantum theory along the structural lines sketched here, it meant a complete and revolutionary structural and semantic departure from all accepted standards in life and science, for studying this world.

Planck has shown that it is impossible to explain the spectral distribution of the energy radiated by a black body under the older assumptions that energy can be divided indefinitely into smaller and smaller parts, but that it may be explained on the structural assumption that the energy exists in quanta of finite size $h\nu$, where ν is the frequency of the radiation and h is a constant ($h = 6.54 \times 10^{-27}$ erg sec.).

These observations lead to the revolutionary structural conclusion that the emission of radiation occurs *discontinuously*, and so the characteristic discrete numbers make their appearance.

It seems natural that because of this peculiar appearance of whole numbers, periodic processes such as rotations or oscillations should be closely related structurally with the quantum theory. As a matter of fact the most important structural and semantic reconciliation of continuous differential equations of the older mechanics with the appearance of discontinuous whole numbers has been solved by the newer quantum mechanics on this basis as explained in the following chapter.

The kinetic theory of heat and the atomistic theory of electricity have shown an enormous productivity. It is quite natural that these theories (verbal structures) should be highly workable, considering the structure of our nervous systems, as explained in the foregoing chapters. It was this principle of individualization which helped so greatly. The quantum theory is a structural attempt to extend this method of individualization, or the atomistic principle, to processes themselves.

As in the older days we introduced units or elementary quanta of mass, and later, an elementary quantum of electric charge, so in our newer knowledge we have need for an elementary quantum of action. Action is defined as energy multiplied by 'time', or $A = Et$.

Naturally such a product as energy multiplied by 'time' must play an extremely important structural and semantic role in this world of space-time,

where nothing happens 'instantaneously', but all action requires 'time'. If we could discover some unit of action, we could change from the language of 'energy' and 'time' to the language of 'action' and 'times'. This language, by the way, is much more satisfactory and structurally closer to experience than the old languages. 'Action' as structurally defined (product of 'energy' by 'time') is one of the two fundamental entities of pre-relativity physics which have survived the Einstein revolution. It is really a universal term which we can apply without danger of degrading science into private gossip. From the neurological standpoint, as it deals with definite units and times, such a term has all the structural earmarks of an abstraction of highest order and of being really semantically important. Energy in space-time must by necessity be reformulated as 'action'. The quantum theory posits structurally that the action of physical processes is built up of a number of elementary quanta of action.

From the fact that electromagnetic waves and light-waves have one velocity, Maxwell concluded that light-waves are of an electromagnetic character, a conclusion which further experiments have fully justified. Einstein, in 1905, successfully applied the quantum structural principle to the theory of light, and in 1907 to the theory of heat of solid bodies.

The evolution of our theories concerning the internal structure of atoms has, until lately, closely followed our astronomical theories, but with the newest quantum mechanics this structural analogy seems less useful. The first atomic model on an electrical basis was proposed by J. J. Thomson. He assumed the atom to consist of a uniformly dense spherical volume charge of positive electricity, within which electrons described circular orbits. But the discovery of radio-activity and the fact that the alpha-rays could pass through several centimetres of atoms (which means penetrating through many thousands of atoms), without their direction being altered made these assumptions structurally untenable.

FIG. 2

The Wilson photographs (Fig. 2) show clearly that a single atom can deflect the alpha-particle by a large angle which makes it clear that the nucleus of an atom must be considered as a very small part of the volume of the atom. The large deflections of the alpha-particles show also that the mass of the nucleus must be much larger than the mass of the deflected particles. Observations show also that the deflections increase with the atomic weight of the deflecting materials. These and similar facts led to the structural assumption that the mass of the atom is principally concentrated in the nucleus and that the mass of the electrons must be very small in comparison with that of the nucleus.

As the atoms are in general electrically neutral, we had to assume that the positive charges of the nucleus are compensated by the negative charges of the

electrons. These and other structural considerations led Rutherford to propose a different atomic model, which was much more successful for a while.

The Rutherford atom is supposed to be composed of a nucleus of positive electricity surrounded by negative electrons. The simplest atom is that of hydrogen, and was assumed to consist of one electron revolving around the simplest positive nucleus or proton, each having a charge, $e = 4.77 \times 10^{-10}$ *e.s.u.*, and different masses, the mass of the nucleus or proton being 1845 times the mass of the electron. The other atoms represent more complex structures built up of protons and electrons, into the details of which we need not enter here. But this theory encountered difficulties of theoretical as well as of experimental character. Niels Bohr eliminated most of them by the application of the quantum theory to the atom.

For simplicity of writing in that which follows, I will use a descriptive language omitting in each statement 'we assume' . , but the reader should be continuously aware, that when we deal with the sub-microscopic levels we deal only with inferential units the representation of which involves a great many assumptions. For my purpose it is enough to stress: (1) the *negative* fact, that the structure of materials is definitely different from that which was assumed before the advent of the quantum theories, (2) that *in science*, inferential units represent abstractions of higher order and are as reliable as the lower order abstractions which we gather on the macroscopic levels, *if* they are treated semantically as *hypothetical* units. The layman should realize that his 'world-outlook' appears as full of assumptions as any scientific one, except that his assumptions are not conscious and cannot be verified, whereas most of the scientific assumptions are conscious and are *continually verified*.

In the older theory the orbits of the electrons were supposed to be arbitrary; in the Bohr theory the orbits have precedence, for which a definite magnitude, a whole number multiple of the elementary quantum of action is specified. We posit one-quantum orbits, two-quantum orbits . , to which definite values of the orbits, the velocity, the number of revolutions, and the energy correspond. In a one-quantum orbit, for instance, the velocity is supposed to be equal to $c/140$, that is one 140th of the velocity of light, and the number of revolutions equal to 6000 billions a second.

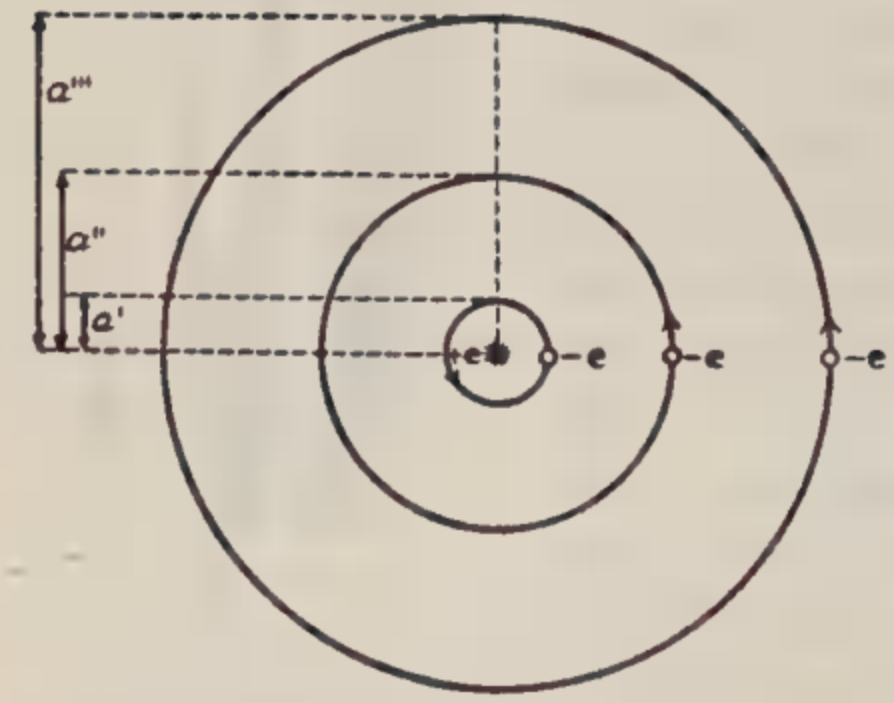

FIG. 3

Bohr later modified his atomic model structurally by taking into account the movement of the nucleus. The electron was not supposed to revolve any longer around the proton, but both proton and electron were assumed to revolve around their common centre of gravity. In the simplest form the Bohr atomic model is shown in Fig. 3, representing the atom of hydrogen, which we assume to consist of a nucleus with one positive

charge and of an electron that revolves about this nucleus. The nucleus is designated by a star, the three circles represent the possible orbits for the electron. The orbit of radius a' is the most stable, and usually the hydrogen electron is supposed to be found there, but through the action of heat or electric fields or collisions . , the electron may be removed to one of the outer orbits a'' or a'''. Such a condition is not so stable, and sooner or later the electron is assumed to return to the orbit a'. During these transitions of the electron, energy is radiated. This structural model is similar to the copernican planetary system, the planet-electrons revolving around the sun-nucleus.[1]

The above diagrams show schematically the supposed structure of some of the simpler atoms. Fig. 4 represents the hydrogen atom, consisting of one proton and one electron revolving around the proton. The mass of the proton is about 1845 times the mass of the electron, and we assume that the proton effectively gives us the mass of the atom.

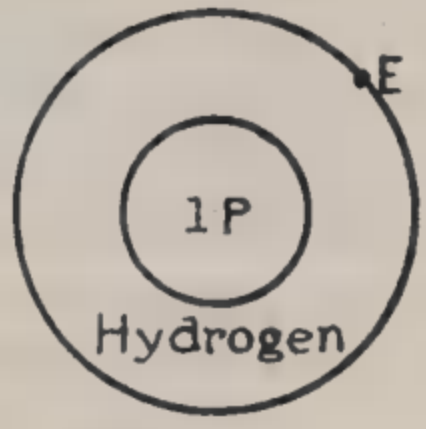

FIG. 4

FIG. 5

Fig. 5 represents the neutral atom of helium. Its nucleus consists of four protons and two electrons, and it has two revolving electrons; in all, four protons with four positive charges and four electrons with four negative charges, the eight charges just neutralizing each other.

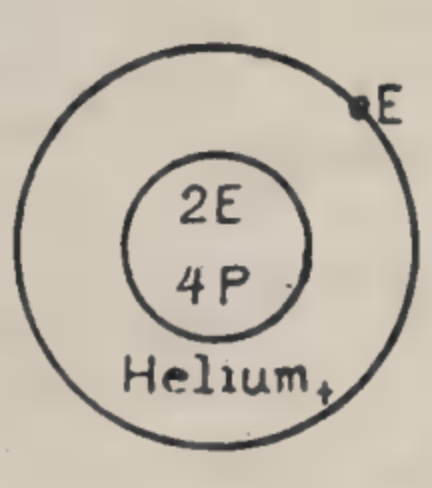

FIG. 6

FIG. 8

Fig. 6 represents a helium atom which has lost one electron. It has, therefore, four positive charges and only three negative charges. Such an atom has a resultant positive charge, and is denoted by He_+. If the helium atom loses two electrons it is doubly charged with a positive charge (He_{++}).

FIG. 7

FIG. 9

P = proton

E = electron

The helium nucleus He_{++}, as shown in Fig. 7, represents the particle emitted from radio-active materials.

Lithium consists of two isotopes; that is, two elements which appear extremely similar to each other in physical and chemical characteristics, but

differ from each other in the number of electrons and protons. In Fig. 8 is shown lithium$_6$, with 6 protons and 3 electrons in the nucleus, and 3 revolving electrons. In Fig. 9 is shown lithium$_7$, with 7 protons and 4 electrons in the nucleus and 3 revolving electrons.[2]

In general terms, Bohr tried to account for all other atoms on the base of the structure of the hydrogen atom. The next important generalization and extension of the Bohr theory was accomplished by Sommerfeld about 1915. The achievement of Sommerfeld can be compared with the advance which Kepler made over the copernican theory of planetary motions. Copernicus considered the planetary orbits as circular. Kepler* considered them as elliptical, and thus introduced a tremendous structural advance in astronomy. Sommerfeld replaced the circular orbits of Bohr by elliptical ones. The theory became much more complicated, because a circle is given by one magnitude; namely, its radius, while an ellipse needs two data, its major and minor axes, and so two quantum numbers for the specification of an orbit. Sommerfeld also introduced some of the results of the Einstein theory; for example, that the mass of a body also depends on its velocity. Since the velocity of the negative electrons in the atom is supposedly very large it was quite probable that the relativity considerations should be appreciable. According to the Einstein theory, the faster a body is moving the greater is its mass. In an elliptical orbit the electron should have a larger mass at the perihelion than at the aphelion, and so the orbit would not be exactly an ellipse but the perihelion would advance slightly at every revolution.

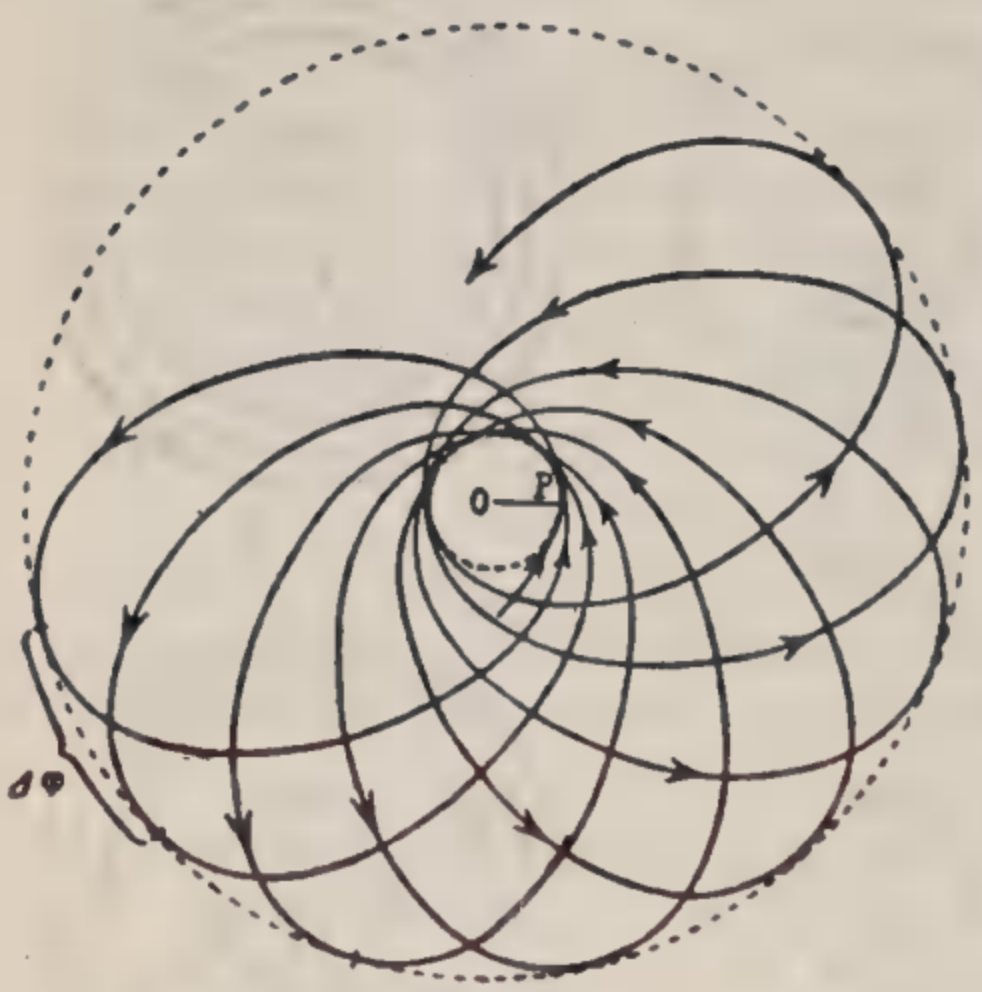

FIG. 10

Fig. 10 gives us the relativistic Kepler orbit as introduced by Sommerfeld. *O* is the fixed focus in which the nucleus is situated and *P* is the initial position of the perihelion. The motion of the perihelion occurs in one sense with that of the orbit.[3]

The last analogy in the structure of the atom taken over from astronomy was introduced in 1925, when Goudsmit and Uhlenbeck proposed their theory of the spinning electron. The electron was supposed to be spinning about its axis like a planet or a top. A similar notion was used by Compton in 1921, in connection with the magneton, but the notion of using the spinning electron for

*Kepler's first law states: 'The planet moves in an ellipse, at one focus of which the sun is situated. Perihelion is that point of a planet's orbit at which it is nearest the sun. Aphelion is that point of a planet's orbit at which it is farthest from the sun.'

the solution of a structural difficulty in the quantum theory, and thus assigning a fourth degree of freedom to the electrons, originated with Goudsmit and Uhlenbeck independently of the work of Compton.[4] It is not necessary for our purpose to follow all the further refinements of the classical theories. Suffice it to say that scientists work under uniquely severe mutual supervision, and that any theories advanced in science are taken under consideration only when the new theories agree better with experiments, and when they also prove structurally fruitful in predicting new experimental facts, which again must stand the test of experiment.

The Sommerfeld orbits have proved to be an advance over the older Bohr orbits, but they had also to be refined to take into account that the electron does not seem to revolve around a simple nucleus but around a *core* consisting of the nucleus and one or more electrons; and so again we had more complex orbits.

For our semantic purpose it is enough to say that to the best of our knowledge (1933), this world appears entirely *different* from what our primitive ancestors knew thousands of years ago, and perhaps from what the average layman knows today. As the problems of 'sanity' represent problems of semantic *adjustment*, and adjustment means adjustment to something—in this case to the structures of the world around and in us—it appears imperative that we should take into account the best knowledge we have of these structures.

The few remarks given above about the structure of 'matter' already show unmistakably that the old 'matter' is not so very 'material', so very 'solid', so very definite, as we once assumed it was; but it represents a *process*. We see that our nervous system, because of its gradual growth and evolution, has developed different levels or strata; our 'knowledge' also has different levels or strata, operative as-a-whole, although different aspects of it can be analysed in terms of order. The reader should realize that because of the old *s.r* we still 'need' some 'bits' of something to speak about. It is a *linguistic* semantic consequence of our pre-scientific, *el* language, which posits absolute 'matter', 'space', and 'time'. Thus, through a process of identification, we ascribe to these terms objective existence. In the old manner of speaking the term 'is' of identity played the main semantic havoc.

In the older days electrons were often taken as 'bits' of something or other. For the layman a 'bit' was identified with 'matter'; and here a great deal of confusion comes to light. Even a 'bit' of something is not necessarily material. Materials, by *definition*, are supposed to exhibit colour, temperature, hardness , . A 'bit' which did not have these characteristics would *not* be material by definition.

Although the 'electron' is defined as an electrical charge, in the older days we had the habit of considering the electrons as some definite 'bits' of something, some kind of 'matter'. Through a process of objectification we made them revolve in definite 'orbits', with definite 'velocities'. , which implies the definite application of *terms* such as 'space' and 'time', derived from *macroscopic gross experience*, but not necessarily applicable to the sub-atomic levels.

What has been said here about the structure of 'matter' is quite sufficient for our purpose. Here, as always, the *negative*—the 'is not'—results count. We are in a position to realize by now that the overwhelming evidence which science gives and which would be impossible to repeat here, shows us a structural picture of the world of tremendous complexity, beauty, and mystery of a structure undreamed of by our primitive ancestors who formulated the current mythological structures which moulded our older *s.r* and languages.

We can sum up, for our purpose, what we know about the structure of 'matter' somewhat as follows. The bits of materials visible and invisible to the unaided eye seem to be less simple than we assume them to be and to the best of our knowledge (1933) represent extremely complex processes of a dynamic structure. It appears also that our usual forms of verbal representation which were built by our primitive ancestors are not similar in structure to the world and so are not fit to represent the happenings going on on the un-speakable levels. As all our knowledge is due to the structure and function of our nervous system, which represents an abstracting mechanism, all our knowledge therefore, appears as some kind of abstractions of different orders, on different levels, of different character, and of varying precision and intensity, resulting in various definite general or individual *s.r.*

To bring what is said here to the lower level of abstracting; namely, to the level of structural visualization and feeling, we may use the rough analogue of an electric or mechanical fan. When such a fan rotates we *see a disk*, simply because our nervous system was evolved under natural conditions necessitating integration, and so does not discriminate between the rotating blades. The separate rotating blades are visually abstracted by us as a single solid disk, although there is no disk present.

To the best of our knowledge atoms represent very minute energetic configurations or dynamic structures where extremely rapid processes are going on, which our nervous system abstracts as 'solid'. Judging by our present standards in science and the amount of knowledge we have we may consider that science in the days of Newton (1643-1727) was in its infancy. During that period we knew a little about the shining specks we see in the skies, and more about the rough macroscopic facts of our daily experience. The genius of Newton not only advanced the detailed knowledge of his day in many branches of science, but also formulated two general theories. One was the differential and integral calculus, which he discovered independently of his contemporary Leibnitz, the other was what we call mechanics.

In Newton's era the problems of macroscopic, microscopic, and sub-microscopic levels of investigation had not yet arisen in the modern sense, although in formulating the differential and integral calculus a theoretical structural step was taken toward the analysis of the processes on the subtler levels. Quite naturally we applied the wisdom we derived from Newton to all phases of life and knowledge. With the advent of more detailed structural knowledge of electromagnetic phenomena which occur on sub-microscopic, as well as macroscopic levels, difficulties began to appear. It seemed as if the

newtonian mechanics were not entirely applicable to these new and smaller scale phenomena. Finally Maxwell (1831-1879) produced his famous theory of electromagnetism. This theory appears structurally at variance with the classical mechanics. Attempts were made to reconcile both kinds of phenomena in one theory. The problems of macroscopic, microscopic, and sub-microscopic structure and levels came to the foreground.

With the advent of the quantum theory further difficulties made their appearance. It became quite obvious that neither the classical 'continuous' mechanics, nor the classical electromagnetic theory could fully account for the 'discontinuous' quantum facts. The situation became acute and bewildering. The Einstein theory with its profound structural semantic and methodological revolution liberated us from our semantic delusions of the uniqueness, absoluteness, and 'objectivity' of 'matter', 'space', and 'time'. It built up a new semantic attitude in the younger generation of scientists already educated on this new structure, and therefore unhampered by the old prejudices. New theories are now being formulated along increasingly more constructive and creative lines.

It is true that as yet neither 'psychologists' nor 'philosophers' have paid enough attention to the subjects discussed here, and so have not made us conscious of the structural and semantic problems involved. However, the Einstein theory has had a profound structural influence on the semantic attitudes of the younger scientists, though in the main they are unconscious of this fact.

The main issues at hand are twofold. One is semantic; namely, to inculcate the permanent structural feeling that words are *not* the things they stand for. If applied habitually, this leads to the rejection of the term 'is' of identity. The other is to replace old languages and methods by structurally new languages and new methods, in which when we describe ordered happenings, we describe the functioning, the behaviour . , by speaking more in a language of what something 'does' than in the old language of what something 'is', which as we have seen *must* be always structurally fallacious and semantically dangerous.

The reader should not take lightly these most general structural and semantic issues. They are unusually important for sanity. When they are formulated we can pass them on, and train children in the new *s.r* quite easily. It is much more difficult, after training a child thoroughly in the *old* vicious structural semantic habits of identification, eventually to have to appoint a guardian angel to watch him day and night to remind him that a word *is not* an object , . Such a procedure would lay a terrific strain on us and on the guardian angel. It would probably be also very expensive, judging by our present earthly substitutes for the 'heavenly powers'.

Once this is realized and applied, the second issue becomes a purely structural linguistic one. There is no *a priori* reason why a language which applies to one level should apply to another.

With these two main issues in view, it is readily understood why modern science tries so hard to develop functional languages and methods in order to be able to describe in terms of order happenings and processes which are ob-

served. Something similar could be said about *all* theories which postulate too much of a definite mechanism, usually involving some identification somewhere.

The slightest discrepancy between such a theory and observation eliminates the theory as structurally unsatisfactory; while theories which succeed in not postulating mechanisms, and so are formulated in a functional language, last much better. One of the enormous advantages of the Maxwell electromagnetic theory is the fact that it describes the behaviour of electricity and magnetism while hardly positing any mechanism at all. A similar statement applies also to the Einstein theory.

The above general remarks are extremely well illustrated by the newer quantum mechanics.

The classical theories, as usual with scientific theories, were very satisfactory in many respects, but not in all; which is an unattainable ideal always demanded from a good scientific theory. They also postulated too much of a definite mechanism, which was the result of, and led to, the semantic disturbance called identification. Indeed, I have read an address by a prominent physicist in which he claims to have 'seen', and invites everybody else to 'see', an 'electron'. He challenges his critics, and seems to feel like fighting—a quite usual result of identification. Electrons represent *inferential entities*, and as such cannot be 'seen', but only inferred, which does not detract at all from the importance of the 'electrons'. The 'seeing' business was good enough in the infancy of science, but not in 1933. We 'see' the stick broken in water, the camera records it as broken, and yet it is not broken. We 'see' the fan as a disk, the camera records it so, but there is no disk. We 'see' a 'solid' piece of wood or stone, which under the microscope proves to have a very different structure , .

In the older days the electrodynamics of moving bodies presented difficulties quite similar to the difficulties encountered in the quantum mechanics. Einstein by an epoch-making stroke of genius solved the problem by observing that, in the languages in question, we operated with a notion of 'simultaneity' which did not correspond to any observable structural phenomenon in the physical world. He discovered that it is impossible to establish the simultaneity of two events occurring at different places, and that a thorough revision of our old theories is necessary in this connection. Einstein formulated a procedure, a method for measurements, taking into account the known laws of the propagation of light and electromagnetic phenomena. He once more established the most important semantic thesis that the laws of nature are relations which are discovered between events which are actually observed, or which are *fundamentally observable.*

It appears that in the older quantum mechanics there were introduced some objectified entities which were never observed, as, for instance, the positions, velocities, and periods of 'electrons' inside the atom. How indeed could we find lengths and 'times' *inside* the atom? Such a procedure requires the introduction of rods and clocks, which themselves consist of atoms; so that *inside* the atom such a procedure cannot be applied. We see clearly that all such conclusions are of an indirect character; but of course such conclusions should

be based on some observable facts, and not only on our freedom to use words in any manner whatsoever. It follows that we must give up a language that speaks in terms of the 'position' of an electron at a given 'time'. , and use instead a language that describes observable characteristics, as, for instance, energy levels which are directly measurable by electron impacts and the frequencies which are derivable from them, the intensity and polarization of the emitted waves . ; instead of electronic 'motions' inside the atom, which never are and never *can be* actually *observed*. It is structurally indispensable to look for such data which are actual or at least can be observed.

As words are not the things we speak about, and structure is the only link between them, structure becomes the only content of knowledge. If we gamble on verbal structures that have no observable empirical structures, such gambling can never give us any structural information about the world. Therefore such verbal structures are structurally obsolete, and if we believe in them, they induce delusions or other semantic disturbances.